PaperSonar

Business requirements

Phase 1 design deliverable v2 | 10 November 2026

Purpose

Users need consistent, inspectable comparisons of candidate research directions, including publication activity, citation context, related datasets/software, creator/institution participation and metadata coverage.

Requirements / design

Functional: ingest five sources with separate state; preserve raw evidence; canonicalize records; retain versions and typed relationships; expose dated metrics and coverage. Non-functional: free-only, replayable, provenance-rich, privacy/security-aware, quota-bounded and restartable.

Scope and caveats

The 20-record OpenAlex sample is only a pilot. Target 30k–60k unique canonical outputs is unverified and depends on profiling.

Source and architecture references

See docs/sources.md, docs/architecture/README.md, docs/requirements/incremental-ingestion-design.md, docs/requirements/course-requirements-matrix.md, and the target architecture diagram. Phase 2 and Phase 3 implementation has not started.